# Inertia of matching cofactor matrices

MANTIS DISCOVERY

MANTIS · MIT CSAIL

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01

## The question

Take a graph $G$ with an even number of vertices and count the perfect matchings that survive deleting two vertices:

$$W_{uv} = \mathrm{pm}(G - u - v), \qquad W_{uu} = 0.$$

How many positive, negative and zero eigenvalues does $W$ have? One positive eigenvalue is the least possible, and it is exactly the Lorentzian property of the polynomial $\frac{1}{2} z^{\mathsf{T}} W z$ in the sense of Brändén and Huh.

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## The five graphs

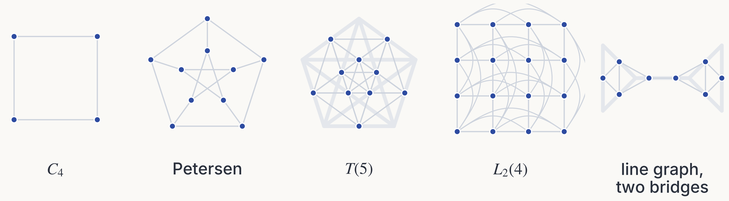



FIGURE 1. $T(5)$ and the bridged graph are line graphs, drawn over their base graphs (faint): $K_5$, and two triangles joined by a path of two bridges. $L_2(4)$ is the rook's graph on a $4 \times 4$ board.

03

## How we are approaching it

- Compute $W$ exactly for many small graphs and look for patterns.
- For line graphs of subcubic graphs, combine a cofactor congruence for one-vertex attachments with the parity of matchings and a signed factorization by two-edge-cut classes.
- For strongly regular graphs, use Gaussian identities and switching for the lattice and triangular families, and exact integer certificates for the rest.

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## Computed spectra

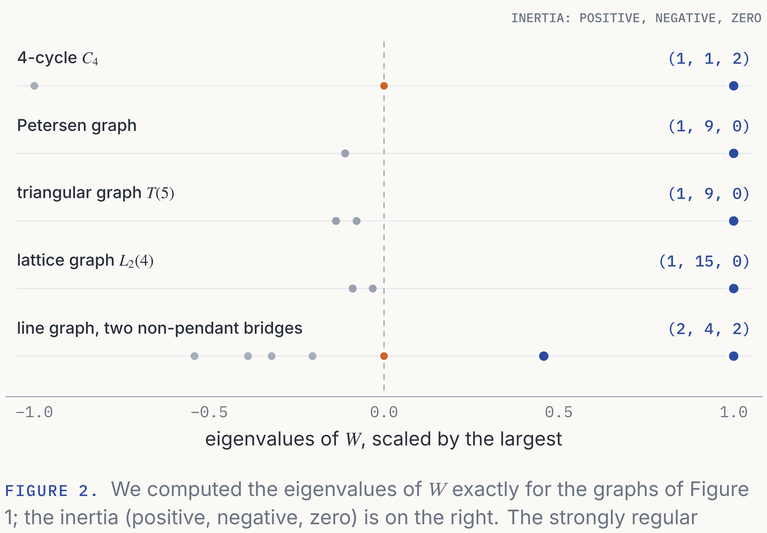



FIGURE 2. We computed the eigenvalues of $W$ exactly for the graphs of Figure 1; the inertia (positive, negative, zero) is on the right. The strongly regular graphs have one positive eigenvalue. $C_4$ is the exception, with two zeros, and a base graph with two non-pendant bridges gives two positive eigenvalues.

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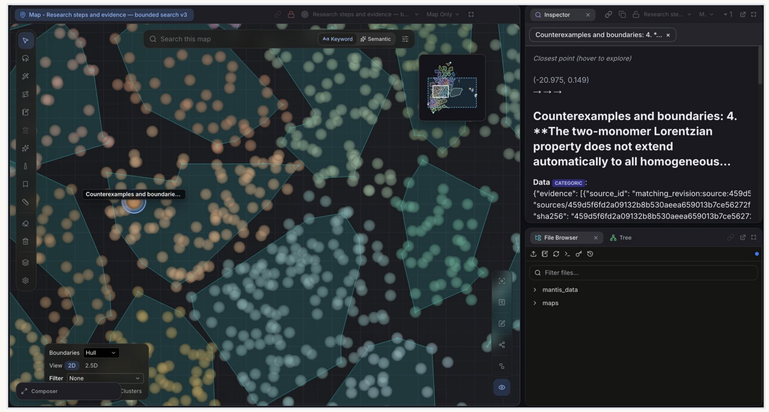
## What we have so far

For line graphs of connected subcubic graphs with an even number of edges, the positive index is a count of components after a rule on bridges. $W$ has one positive eigenvalue exactly when every bridge of the base graph is pendant.

Every connected strongly regular graph of even order with least eigenvalue $-2$ has inertia $(1, n-1, 0)$, except $C_4$, which has $(1, 1, 2)$.

06

## In Mantis




3,583 RECORDS · THE OPEN RECORD

*The two-monomer Lorentzian property does not extend automatically to all homogeneous polynomials.*

The record gives an example: for an eight-vertex cubic base graph, the four-monomer polynomial is not Lorentzian.

ALSO ON RECORD

An invalid proof step assumed that homogeneous components of real-stable polynomials are real-stable. They need not be: $z^2 - 1$ is real-stable, but $z^2 - t^2$ vanishes at $z = t = i$.

07

## Open questions

- Which deletion-count matrices are Lorentzian, and what makes the others fail?
- Line graphs with least eigenvalue $-2$ can have an arbitrarily large positive index. What controls it in general?